BORA

4 SEPTEMBER 2025

STRUCTURAL SYSTEM EVALUATION PROPOSED PROCESS AND TIMELINE FOR THE JEFFERSON AND IDA B WELLS HIGH SCHOOL MODERNIZATION PROJECTS.

Following the Board of Education framework within the topic of implementation of PPS's Climate Crisis Response Policy, the Board directed OSM to "evaluate structural materials choices such as mass timber versus steel, weighing factors such as embodied carbon, durability and cost". To assist with this evaluation, Bora suggested the following process for the development and evaluation of the structural systems for both the Ida B Wells and Jefferson High School projects.

PROCESS FOR RESEARCH AND EVALUATION

Step 1: Research, Data, and Examples

During the 2024 design efforts, Bora began compiling a list of potential concerns from PPS for the use of Mass Timber so the project teams could look for solutions or mitigation strategies. On May 9th, 2025, PPS sent additional notes for the teams to consider. These notes and comments will ensure we design a school facility that meets PPS's goals for carbon emission reduction, supports student outcomes, and results in a facility that is maintainable with the extremely limited budgets and staff available.

Specific questions from PPS sent to the design team include:

- Exposed mass timber ceilings (avoidance of clutter and dust surfaces, ceiling heights and devices, acoustics, etc.)
- Repairability and maintenance of wood surfaces (vandalism, small fire, general wear, etc)
- Planning flexibility
- Insurance: Builder's risk costs & ongoing insurability of combustible buildings
- Embodied carbon comparison between structural materials
- Data regarding student outcomes and student/staff health impacts

While the project teams had several ideas and design solutions in process that address some of areas of discussion, the teams held off on further research on the above topics until it was known if the Bond would pass. With its passage, the Bora teams and our consultants resumed collecting data and research and iterating design ideas to address questions from PPS, suggesting solutions where possible, and understanding costs and benefits. In addition, Bora connected with other school districts that have used mass timber to understand their maintenance and durability experiences and shared contact information with OSM to facilitate conversations with PPS. For example, Seattle Public Schools has two completed mass timber schools and one in construction. Bora and representatives from OSM have also visited Lakeridge Middle School, which has some selected mass timber including exposed columns. Also, Bora met with Lincoln High School's chief custodian to discuss maintenance issues at that facility and compiles notes regarding wood elements in that school. Lastly, Bora's data included clarification of certain areas that are often not well understood, such as potential ceiling strategies where mass timber is proposed. All in all, the design teams worked to both progress the designs while concurrently considering maintenance and cost for the Schematic Design pricing sets issued in August.

During July and August, the Design Teams prepared (4) update sessions for the OSM team addressing specific areas, as follows:

Session #1: Alignment with Values

- Summary:
    - Significant data shows that mass timber is part of a trauma-informed design strategy with biophilic attributes. This means that there are physiological and mental health benefits to the presence of exposed wood, in addition to emerging data in improvements in test scores, focus, creativity, and productivity.
    - Mass timber has significantly better performance in carbon reduction, part of the PPS Climate Crisis Response Policy.
    - Generally, people consistently describe spaces with exposed wood as more welcoming and inviting. The use of exposed wood improves the student and staff experience in those areas.

- Presentation: https://boraarchitects.sharepoint.com/:b:/t/30210IdaBWellsHSModernization/EZU1XkOVCkZNl2J9QQtVrncBh8ScS2NHVSGiewhww8Qd-A?e=bf8amn
- Recording link: https://bora.zoom.us/rec/share/fbh3tLoC3EgxAc-j-aBpxpL7jDkHmJkOjbinSs7Gym-u_F43t8GBMt0zmRb7MgPo.0Npe48FH-z2KdeIT
  Passcode: .yKux6nY
- Mithun Report (referenced in the slides): MassTimberSchools_Report.pdf

Session #2: Maintenance & Repair [Touch zone]

- Summary:
    - While “mass timber” is a relatively new material for PPS, mass timber and heavy timber are not a new material in general for use in institutional buildings for which durability and longevity are critical.
    - Mass timber is as repairable as concrete or steel for uncommon damage that might occur from significant fire or long exposure to water.
    - While other Districts with exposed mass timber have not experience vandalism on the wood, the team explored strategies to prevent, protect, clean and repair and wear.
    - From a comparison activity, the team found that wood is more visually forgiving than drywall for minor scratches and dings, and wood is durable, cleanable, easily repairable.

- Presentation: https://boraarchitects.sharepoint.com/:b:/t/30210IdaBWellsHSModernization/EQetmKHle1JIt-b0-sWPhsoB0TaMKx868_5m5usEW1qLGg?e=Q8KiwK
- Recording link: https://bora.zoom.us/rec/share/F5ZBhYc6HF-VR29WWC1b31O9e9y-bn3nkpxWUvaD1XcV86wQyJIeduKAwDuf6SI.v7lKMRxi-ST1Nhv9
  Passcode: +q85Czr#

Session #3: Ceilings & Acoustics [Overhead zone]

- Summary:
    - The design goal for all ceilings is to have clean, clutter-free ceilings that allow for easy maintenance and access. Properly designed and coordinated mass timber ceilings
        - should not have distracting clutter, dust or debris.
        - allow for accessible and visible systems for maintenance, replacement, and repurposing.
    - Exposed wood in classrooms create less “institutional”, warm, and “biophilic” learning environments.
    - Timber ceilings have a range of acoustic options that need to be evaluated to choose the best solution for PPS.
    - The additional coordination of systems needed for mass timber during design has benefits to avoid conflicts during construction.

- Presentation: https://boraarchitects.sharepoint.com/:b:/t/30210IdaBWellsHSModernization/EXLT1Xc6h0FBlQQPJPriNsYBkU8njeJfZvNd8Id4uX8-8g?e=sz595d
- Recording link: https://bora.zoom.us/rec/share/3lYcSaSmp3MxpoNqn0cpLkzprxJQdSEg6kuCkLl1YyPP0VjQL7xqhiclCdrUbO-8.s82drbYxU9xBdGQG
  Passcode: ALJ5yM%h

Session #4: Insurance, Risks, and Evaluation Criteria

- Summary:
  - Builders Risk Insurance costs can vary greatly between insurers, but a recent builders risk quote that Hoffman obtained from Zurich insurance indicate only a $300,000 delta between Steel and Timber options for the duration of the IBW modernization construction.
  - Insurers pay particular attention to risks and mitigation measures being employed, including team familiarity with timber, package value and duration, fire and water mitigation during construction, and overall collaboration and communication with the owner, construction, and design team.
  - We would not expect any added ongoing property insurance premiums for mass timber buildings as opposed to a steel building.
  - Both steel and timber are experiencing global supply chain and market volatility, whether sourced domestically or internationally. The current designs use mass timber in only the classroom wings of the projects, and steel in the athletics and performing arts wings, hedging against sudden increases in one of these.
- Presentation: https://boraarchitects.sharepoint.com/:b:/t/30210IdaBWellsHSModernization/EeSYv_o2Cu9PvI4mgp-ovgIBkjjsFETq7FZNpfUonvzlGQ?e=Kt4UGE
- Recording link: https://bora.zoom.us/rec/share/jFTcx-8QKIeD_YwuBONm2W3icAyHvUl726bswya3O0sBYtE9aTkuY1H8GrCOdMQL.DF4a14L7J4Wx5BVW
  Passcode: .C5G$kG^

Step 2: “Mass Timber Summit” - Data Review Discussion

Incorporating feedback from OSM on the individual sessions above, Bora prepared for a larger presentation – “Mass Timber Summit” – to PPS Operations and Facilities staff, many of whom will be maintaining and operating the two high schools. The Summit was held on September 3rd, 9am – 12pm. The goal for this Summit was to introduce mass timber to those who are less familiar with mass timber, to address the questions and concerns shared in the May 9th memo, and to uncover if there are additional areas of concern or questions. The slides for this presentation were made and condensed version of the four previous sessions with OSM suggested updates.

Presentation:
https://boraarchitects.sharepoint.com/:b:/t/30210IdaBWellsHSModernization/EWHDrN6hkTVCvlj4Vf7iHoUBCGyMWLihjBkgLQCzfmIDnQ?e=gt6KYc

Structural Evaluation Criteria and Matrix

Superintendent Dr. Armstrong requested that the OSM team present recommendations on structural systems along with the Comprehensive Plan updates scheduled to be presented at the Facilities Improvement Oversight Committee meeting on September 16th. Even though the timing is ahead of receiving cost information from the Schematic Design issuances, because there are two examples that show that mass timber can be relatively cost neutral with a steel building, the assumption for this structural evaluation matrix is that this will be true for these two high schools as well, given the time for the design and General Contractor teams to do this work.

In order to evaluate the structural system, OSM has suggested that the model for this document could be the data evaluation matrix used to evaluate the structure for the earlier iteration of Jefferson High School. In parallel with the Mass Timber Summit, OSM and Bora worked collaboratively to update previously used criteria to evaluate the advantages of mass timber or steel building systems for PPS. The matrix below shows the criteria used and the definition of the criteria for evaluation. By using the colors in the scoring key to indicate positive and negative impact of each system for each criterion, narratives describe why the impact score was given.

While OSM and the design team considered different ways to weight criteria, ultimately, the assessment of positive or negative impact alone was sufficient to show reason to recommend mass timber for the classroom wings of both high schools.

STRUCTURAL SYSTEM EVALUATION WORKSHEET

THE PURPOSE OF THIS WORKSHEET IS TO HELP EVALUATE STRUCTURAL OPTIONS BEING ANALYZED BY THE JHS + IBW HS MODERNIZATIONS TEAMS.

CRITERIA	DEFINITION	MASS TIMBER HYBRID BUILDING	STEEL FRAMED BUILDING	SCORING KEY
CONSTRUCTION SCHEDULE + LOGISTICS	The baseline schedule is built around a mass timber hybrid structure. Impacts are relative to better or worse than this baseline for project schedule, construction staging and logistics, construction complexity, and procurement of long-lead materials. RELATES TO: impact on cost, PPS oversight, on-site safety and disruption	fastest to construct. BOD schedule for project. key areas of why so fast: timber frame erects faster than other structures, no composite decking, no concrete above grade, no additional finishes/trades following erection.	slower to construct 3-4 month longer duration than timber. key differences to timber structure: composite concrete decks and fireproofing of steel; additional finishes and trades later in project.	MAJOR POSITIVE IMPACT +10 points
MAINTENANCE / REPAIR + DURABILITY	Estimated relative need for protection or maintenance, the relative ease or need for repair or replacement, and the general longevity of materials. RELATES TO: PPS ongoing operations and limited Facilities resources	Most wood columns will be exposed in the touch zone but sealed with multi-layer clear finish which will prevent absorption into the wood. Most marks, scratches, and gouges will not be highly visible and may not need any repair. Cleaning: clear finish surfaces can be wiped down. Matching of refinished surfaces can be challenging, but is often not necessary. Repair is likely to be apparent but may not be objectionable. Biggest challenge is corridor floors above grade are not concrete. Depending on finish material, this may require yearly maintenance.	All structure in public spaces to be covered with painted gypsum board. While most surfaces can be easily patches and painted, marks, scratches and gouges will be highly visible, detracting from appearance, and needing PPS maintenance to repair correct. Biggest challenges: Matching of repainted surfaces can be very successful, but colors other than standard white may not be in PPS stock. More difficult to not require surfaces; or, leaving marks will detract from appearance long term.	
FUTURE FLEXIBILITY	Potential for spaces to be *repurposed* and accommodate additions for growth. While wall reconfiguration is less common, modification of building systems (power, data, plumbing) to change space usage is more common. This also includes future penetrations through floors or beam. RELATED TO: PPS long term operations, limited Facilities resources, and adaptability	15'x30' Grid Spacing with 1-way frame Denser and larger column grid may make repurposing a space more challenging. 1-way structure allows for routing of pipes and conduit easily from the main corridor, but is more difficult across beams. Holes through floor are easily accommodates, but requires attention to gypcrete, acoustimat, and floor finish.	30'x30' Grid Space with 2-way frame fewer columns may allow for more flexibility in repurposing. 2-way structure can have predefined holes for routing future pipes and conduits, but in limited locations. Holes through the floor need to have care to avoid rebar, but concrete patch is relatively simple.	MINOR POSITIVE IMPACT +5 points
OCCUPANT HEALTH + EDUCATION OUTCOMES	Potential impact on student and staff physical and mental health and absenteeism. Potential impact on student outcomes, such as test scores and sense of belonging. RELATED TO: PPS mission for student outcomes, long term operations (staff, teacher, and substitute teacher costs)	Visual presence of wood contributes to biophilic and trauma-informed design, which improves student and staff health, student test scores, and sense of belonging without added finishes. Wood sealers are low-to no-VOC.	The structural steel frame requires sprayed-on fireproofing which contains carcinogens, but which are generally encapsulated and not exposed except during installation; this is only a minor negative in that needing to encapsulate steel does not provide any benefits to health or outcomes.	
CLIMATE POLICY	Potential ability to meet PPS's goals as stated in the Climate Crisis Response Policy, specifically in terms of embodied carbon, recyclability, and reusability. RELATED TO: PPS Climate Crisis Response Policy and PPS values in community	Utilization of mass timber structure in the classrooms dramatically reduces the school's embodied carbon. In addition, mass timber can be recycled and reused with significantly less energy than steel, and without the waste of added fireproofing, drywall, etc.	Utilization of a structural steel frame, including cementious fireproofing and added gypsum surfaces dramatically increases the school's embodied carbon. In addition, while steel can be recycled, this requires significant energy. The added framing, drywall, and paint go into landfills.	MINOR NEGATIVE IMPACT -5 points
APPEARANCE + FINISHES	Relative quantity of different finishes that are required for initial installation (trades, time) and later storage (maintenance impact). Relative opportunities for design to promote quality and community connection, and to inspire inhabitants. Also considers ability to contribute to daylit and aesthetically pleasing spaces. RELATED TO: project costs, limited Facilities resources, PPS mission for students and community; and PPS Climate Policy	Substantially fewer additional finishes required. Sealers are only needed in the touchzone and ends.	Structure needs to be entirely covered with applied finishes for fire rating, occupant safety, and appearance. This includes framing in columns below the ceiling and covering ceilings entirely in acoustic tile or drywall. This requires later maintenance.	
SUPPORT LOCAL INDUSTRY + EQUITY	Potential ability to support local or regional industries, companies, and crafts, as well as to employ or enhance equity in contracting goals. Could also include transparency of supply chain to avoid objectionable practices such as slave labor, inhumane practices, or cultural degradation. RELATED TO: PPS values to support community	Potential for wood sourcing to support local Pacific Northwest industries, including partnering with local tribal sources. As seen at PDX, potential to support local small businesses and advance PPS equity in contracting goals. Wood for mass timber in this area comes from sources without slave labor or inhumane practices.	There are not local sources of structural steel members; as such procurement of structural steel would come from outside the region.	MAJOR NEGATIVE IMPACT -10 points
DESIGN + COORDINATION EFFORT	Estimated effort required by the design and preconstruction team to develop a fully coordinated design that integrates systems, in particular the overhead mechanical, electrical, and low voltages systems. Note that additional effort in design may be offset by less coordination needed during construction. Also considers system's ability to remain exposed to view for ease of maintenance. RELATED TO: project costs/process and limited Facilities resources	Exposed structure requires careful coordination and detailing during preconstruction to ensure desired final outcome. However, lessons learned show that this effort is largely offset significant construction-phase coordination savings since all the work is done early. Conduit cannot be located within the slab.	Less coordination is required initially for systems hidden above the ceiling; however later coordination during construction is still necessary. Conduit can be located in the concrete slabs if necessary.	
CONSTRUCTION DISRUPTION	Potential ability for off-site construction or prefabrication in order to reduce on-site noise, material storage, and construction waste, and therefore reduce trucks and workers on site. RELATED TO: on-site safety and student/staff disruption as well as PPS value to support community (minimize impact to neighbors)	Mass timber is highly conducive to prefabrication, which significantly reduces noise on-site, as well as the number of workers needed to erect the timber frame.	Structural Steel would not reduce distruption on site.	EXTREME NEGATIVE IMPACT -20 points
CONSTRUCTION COSTS	The cost of holistic construction, including materials, builder's risk insurance, labor pool.	For this evaluation, the assumption is that costs will be relatively neutral, based on previous work and CHS estimates.	For this evaluation, the assumption is that costs will be relatively neutral, based on previous work and CHS estimates.	NOT SIGNIFICANT DIFFERENTIATOR FOR THIS

Step 3: Cost Estimating and Reconciliation

While the teams worked through Schematic Design phase to optimize costs for both mass timber and steel systems, the design team expects further ideas for optimization to emerge during the cost estimate review. In addition, previous experience indicates that several areas of optimization and understanding occur during the Design Development phase, and these areas of detail are critical to finding the best way for mass timber to be relatively cost neutral with a conventional steel building.

Note: The Ida B Wells HS team will be working with Hoffman Construction prior to the estimating process; however, the Jefferson HS team did not have access to a General Contractor during most of the Schematic Design Phase. The Colas-Hoffman team was onboarded during the final three weeks of Schematic Design.

The Ida B Wells HS team hopes to have a reasonable comparison of the holistic but initial costs of a building using a mass timber structure and one using conventional steel following the third-party VE workshop scheduled for the week of September 15th. The Wells team plans to hold a Schematic Estimate Review the week of September 22nd.

The Jefferson HS team will not have initial costs until the week of October 13th. While the JHS team will not hold another third-party VE Workshop, the team will be considering path to budget and cost savings options (if needed) at that time.

Both teams expect to continue working towards further detail and optimization through 25% Design Development. However, the recommendation for the classroom wing to be mass timber will allow the design and construction teams to focus on one system.

TIMELINE FOR RESEARCH AND EVALUATION

Step 1: Research, Data, and Examples through end of August with the completion of 100% SD sets

- 8/15/25 - IBW 100% SD Issuance
- 8/27/25 - JHS 100% SD Issuance

Step 2: Data Review Discussion & Work Session

- Week of 9/2/25 - IBW/JHS Mass Timber Summit with Facilities & Operations

Step 3: Cost Estimating and Reconciliation following completion of Cost Estimating process.

- Week of 9/22/25 – IBW SD Cost Review
- Week of 10/13/25 – JHS SD Cost Review